

The locked geometry panel, and why each piece is in it

This thesis extracts five things from every coronary centerline: three curvature-based numbers and two decisions about how those numbers, and the local curvature they come from, are computed and kept. All five are locked in here because each has an evidenced or a mechanistically argued connection to wall shear stress (WSS), the frictional drag blood exerts on the artery wall as it flows past. That drag is not uniform along a vessel; it falls, or reverses over the heartbeat, wherever the vessel bends or divides, and the endothelial cells lining the wall respond to chronically low drag by turning inflamed and plaque-prone, which is where and why plaque starts [1]. Tracing the flow through human coronary trees after death, Asakura and Karino found almost every plaque in exactly two places: the inner wall of a bend and the outer wall just past a bifurcation [2]. The five measures below exist to locate those two places without running a flow simulation for every patient.

1. Mean absolute curvature, the primary measure

Curvature κ describes how sharply a curve bends at a point: a tight bend has a small radius of curvature R and a large $\kappa = 1/R$, and a straight stretch has $\kappa = 0$. A vessel's curvature therefore changes continuously along its length, so the first locked measure takes its average over the whole vessel,

$$\bar{\kappa}_a = \frac{1}{L} \int_0^L |\kappa(s)| ds, \quad (1)$$

where L is the vessel's length and s the distance traveled along it. Every local bend adds positively to this average, because the absolute value keeps a bend to the left from canceling a bend to the right; the same quantity can be computed by summing the turning angle between every pair of consecutive short steps along the vessel and dividing by L , a construction called the sum of angles metric [3].

Kashyap and colleagues tested this measure, and three others, against a hemodynamic simulation of 127 people without coronary artery disease: for each person's left main bifurcation, they computed the percentage of the vessel wall running below 0.4 Pa of time-averaged wall shear stress, a simulation's stand-in for a chronically low-shear zone, and regressed each candidate tortuosity measure against it [4]. Mean absolute curvature $\bar{\kappa}_a$ (Equation (1)) explained the most variance of any measure they tried, $R^2 = 0.112$ ($p < 0.001$). The classic **tortuosity index**, the ratio of the vessel's arc length to the straight-line distance between its ends, explained essentially none, $R^2 \approx 0$ ($p = 0.865$): a vessel can be long relative to that straight line either because it winds back and forth or because it sweeps through one long, gentle curve, and the ratio cannot tell those two apart. This is why $\bar{\kappa}_a$, not the tortuosity index, is this thesis's primary locked measure, and why the tortuosity index is not carried forward as a feature at all.

2. Root-mean-square curvature, the sharp-bend companion

A single sharp kink and a long, gentle arc can share the same total turning and so the same $\bar{\kappa}_a$, yet a sharp kink disturbs flow more. The second locked measure weights sharper bends more heavily by squaring curvature before averaging,

$$\kappa_{\text{rms}} = \left(\frac{1}{L} \int_0^L \kappa(s)^2 ds \right)^{1/2}. \quad (2)$$

In the same 127-person test, κ_{rms} also tracked the low-shear area, $R^2 = 0.093$ ($p = 0.002$) [4], close to $\bar{\kappa}_a$ though not quite as strongly. The same quantity, defined for retinal vessels as total squared

curvature normalized by length, was Hart and colleagues’ original tortuosity measure, and separated tortuous from normal retinal vessels with 91 to 95 percent accuracy [5] (read at abstract level only; the size of that classification test is not recorded here and should be checked before this note is merged into the thesis). Because it squares a derivative, κ_{rms} is noisier than $\bar{\kappa}_a$ and moves more with how much the centerline is smoothed before differentiating, so it is reported only alongside a stated smoothing scale, never as a bare number.

3. Torsion and non-planarity, the out-of-plane part

Curvature alone cannot tell a bend that stays flat from one that also twists out of its own plane, the way a coiled spring twists as it bends. **Torsion** τ is the rate at which the plane of the bend rotates about the vessel’s own direction of travel: zero for a bend confined to one plane, and nonzero once the vessel corkscrews out of it. Some simulation studies argue that this out-of-plane twisting drives a spiral, helical flow pattern that protects the wall rather than harming it [6], though that claim is read here at review level, from a table summarizing other groups’ simulations, not from the primary studies themselves. Because torsion might help rather than hurt, it is kept as its own number rather than folded into $\bar{\kappa}_a$ or κ_{rms} , so a vessel that twists a great deal is never mistaken for one that simply bends a great deal.

4. Bifurcation angle together with branch diameter

At a bifurcation, the angle between the two daughter vessels changes where the incoming flow separates from the wall, but the angle alone does not say how much flow each daughter carries: that is set mostly by each daughter’s own diameter. Pooling 1070 coronary trees, Taylor and colleagues placed this flow-to-diameter relation, known as Murray’s law, at an exponent of 2.39 (95% CI 2.24 to 2.54), close to but below the cubic value classical theory predicts [7]. Because the drag a branch’s wall feels depends steeply on how much flow it carries, two bifurcations with the same angle but different diameter ratios load their walls differently. The fourth locked measure is therefore the angle and the two daughter diameters together, extracted as one unit, rather than the angle on its own.

5. Keeping curvature as a profile, not one number per vessel

The first three measures each reduce a whole vessel to a single number, and a single number can hide where along the vessel the bending actually sits. Tello Ayala and colleagues trained a model to read the turning-angle profile along each vessel directly, rather than its average, on 22,334 patients undergoing angiography, and it predicted coronary artery disease better than the same information collapsed to its global mean, an area under the receiver operating characteristic curve (AUROC) of 0.67 against 0.60 for the mean alone [8] (read at abstract level only). The fifth locked choice follows from this: this thesis keeps the curvature profile $\kappa(s)$ along each vessel, per segment, rather than collapsing it to $\bar{\kappa}_a$ immediately, so that where a bend sits is never thrown away before it can be used.

Considered, not locked

Three further ways of scoring tortuosity were considered and left out. Grisan and colleagues’ tortuosity density cuts a vessel at each point it changes bending direction and combines the arc-to-chord ratio of every resulting piece [9]; it captures turning as well as elongation, but it inherits whatever noise the inflection-point detection carries, on top of the noise curvature already carries, for

no clear gain over $\bar{\kappa}_a$ on the one evidenced test above. Categorical clinical scores, such as counting bends sharper than 45 degrees or consecutive 180-degree turns [10], are simple to read off a scan but throw away the continuous magnitude that $\bar{\kappa}_a$ and κ_{rms} keep, and neither has been tested against a shear simulation the way the measures above have. And a single tortuosity score per patient, rather than per vessel segment, was ruled out directly: across 1010 people undergoing angiography, more tortuous arteries were, if anything, associated with less coronary artery disease, not more [11], the opposite of what a whole-patient score implies if disease tracked tortuosity everywhere. Flow acts locally, so this thesis measures locally, one vessel segment at a time.

None of the five locked measures is a direct measurement of wall shear stress, and a high score on one is a candidate risk marker, not proof of risk. Even a full hemodynamic simulation only moderately separated lesions that went on to cause a heart attack from those that did not, in patients who already had one, area under the curve 0.63 to 0.68 [12]; a shape measure computed without running that simulation should not be expected to do better.

References

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correlation with low shear ($p = 0.86$). This is the source for objective 7 computing average absolute curvature rather than a tortuosity index.

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105(4):14–17, 2009. [BIBLIOGRAPHY VERIFIED] PMID 19585899; no DOI issued. n = 1221 consecutive coronary angiograms, severe tortuosity in 12.45%, associated with a significantly LOWER incidence of significant CAD ($p = 0.003$). The largest tortuosity cohort found; author given names are PubMed initials.

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